

Sharp relative balance at the endpoint

Complete proof; independently reviewed

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Assume positive row weights w and nondecreasing columns $p_i \in [0, 1]$, $i \in [m]$, satisfy $\mathbb{E}_w \prod_{i \in S} p_i = 1/(|S| + 1)$ for every subset S . Put $u_i = 1 - p_i$ and $s = \sum_i u_i$.

Theorem 1. *There is an absolute constant C such that every row satisfies*

$$\max_i u_i \leq \frac{C}{m}(s + m^{-1}).$$

The order m^{-1} of the additive term cannot be replaced by $o(m^{-1})$ uniformly in this tensor class; see `endpoint_profile_sharpness.tex`.

This is a consequence of the reviewed quadratic-bound machinery, not an input to its proof. We use its notation $\ell = \lfloor m/4 \rfloor$, $\alpha = \ell/m$, $x = \alpha s$, $\bar{A} = \mathbb{E}_B \prod_{j \in B} p_j$, and the measures $\mu = (\ell + 1)w\bar{A}$, $\nu = mw$. The scalar target is $b_0(z) = (\ell + 1)\ell^{-1}(1 - z/\ell)^\ell$ on $[0, \ell]$. Set $\theta = 1/16$ and

$$T_d(y) = \sum_{j < d} (1 - j/d)L_j(y), \quad U_d(z) = d^{-2}T_d(\theta z)^2.$$

All constants below may depend on the fixed absolute constants in already established estimates. Only two further iterations are used.

Lemma 1 (A sharp endpoint distribution bound). *For every such tensor,*

$$\nu([0, t]) \leq C(t + m^{-1}) \quad (t \geq 0).$$

Proof. The proof of the U_d comparison in `size_bounds/quadratic_per_die_lower_bound.tex` is uniform for $m^{1/5} \leq d \leq \kappa_0 m$, with $\kappa_0 > 0$ sufficiently small. Indeed its small, middle, and large regions give, with $\delta = m^{-1/3}$ and $h = m^{-3/4}$,

$$|\mu(U_d) - \beta_0(U_d)| \leq C \left\{ \frac{\log d}{dm} + \frac{h}{m} + \frac{\delta}{m} + \frac{\delta h d}{m} \right\} + O(m^{-3}). \quad (1)$$

To check the uniformity, the small-region contribution is $C(d/m)(1/d + \delta)(1/d + h)$, bounded by the displayed terms. The good middle-region envelope is $C(dm)^{-1}e^{-cx}(x^{-1} + \delta x^{-2})$. The bad-subset exponent is $-cmx/(x + \delta)$ for $x > 1/d$; it dominates every polynomial factor throughout this degree range. The large-region tail has a polynomial factor times e^{-cd} . These are exactly the estimates used in the cited proof, with the same choice of a small upper bound on d/m .

Multiplying (1) by d gives $o(1)$ uniformly: the largest term is $\delta h d^2/m \leq \kappa_0^2 m^{-1/12}$. Thus $\mu(U_d) \leq C/d$, since $\beta_0(U_d) \leq C/d$. Also $U_d \geq c$ on $[0, c_0/d]$, and $\nu \leq C\mu$ there because $\bar{A} \geq 1 - s \geq 1/2$. Choose d comparable to $\min(\kappa_0 m, 1/t)$ whenever $t \lesssim m^{-1/5}$. The previous bound $\nu([0, t]) \leq C(t + m^{-3/4})$ handles larger t . This proves the assertion, including a possible atom at zero. $\square$

We next state a quantitative improvement mechanism. Suppose that, uniformly for tensors of every sufficiently large dimension m ,

$$\max_i u_i \leq C_*(s + \delta_m)/m, \quad \delta_m = m^{-a}, \quad 1/3 \leq a \leq 1. \quad (1)$$

The constant C_* is fixed. Deleting one column gives the same hypothesis with dimension $M = m - 1$, and $\delta_M \asymp \delta_m$. For a fixed deleted index i , let $s_- = \sum_{j \neq i} u_j$, $\ell = \lfloor M/4 \rfloor$, $x = (\ell/M)s_-$ and let $\bar{A}$ average the success products over ℓ -subsets of these M columns. Define

$$\mu_0 = (\ell + 1)w\bar{A}, \quad \mu_1 = (\ell + 1)(\ell + 2)wu_i\bar{A}, \quad \nu = Mw.$$

These are probability measures except that ν has mass M . Their scalar targets satisfy $b_1(z) = ((\ell + 2)/\ell)zb_0(z)$, and $d\mu_1 = (\ell + 2)u_i d\mu_0$. Absorbing the u_i term in (1) gives

$$Mu_i \leq C(s_- + \delta), \quad \delta = M^{-a}. \quad (2)$$

Lemma 2 (Marked taper comparison). *There is a sufficiently small fixed $\kappa_0 > 0$ such that, uniformly for $M^{1/5} \leq d \leq \kappa_0 M$,*

$$|\mu_0(U_d) - \beta_0(U_d)| \leq C \left\{ \frac{\log d}{dM} + \frac{1}{M^2} + \frac{\delta}{M} + \frac{\delta d}{M^2} \right\} + O(M^{-3}), \quad (2)$$

$$|\mu_1(U_d) - \beta_1(U_d)| \leq C \left\{ \frac{1 + \delta \log d}{dM} + \frac{\delta^2}{M} \right\} + O(M^{-3}). \quad (3)$$

In particular $\mu_0(U_d) \geq c/d$ uniformly in this range for large M , and

$$\mu_1(U_d) \leq C/d^2 \quad \text{if additionally } d \leq \sqrt{M}/\delta. \quad (4)$$

Proof. The exact marked polarization identity is proved in `rational_designs/asymptotic/quantitative_relative_endpoint_balance.tex`. The mark is outside all success and polarized subsets, so the same subset moments and Taylor estimates apply. The proof of (2) is the unmarked proof recalled in Lemma 1, now using $h = M^{-1}$ from that lemma and the bound (1).

For (3), the extra normalized marked factor is at most $C(x + \delta)$ by (2). In the small region $x \leq 1/d$, the integrated comparison error is at most

$$C \frac{d}{M} (1/d + \delta)^2 \nu([0, 1/d]) \leq \frac{C}{M} (1/d + \delta)^2,$$

because $d \leq \kappa_0 M$. In the good middle region the unmarked envelope therefore becomes

$$\frac{C}{dM} e^{-cx} (1 + 2\delta/x + \delta^2/x^2).$$

The sharp CDF bound gives

$$\int_{1/d}^{\infty} e^{-cx} d\nu \leq C, \quad \int_{1/d}^{\infty} e^{-cx} x^{-1} d\nu \leq C(\log d + d/M),$$

and

$$\int_{1/d}^{\infty} e^{-cx} x^{-2} d\nu \leq C(d + d^2/M).$$

Substitution proves (3). The bad-subset and large-region errors remain $O(M^{-3})$: the extra marked factor is bounded by a polynomial in M , while their bounds are superpolynomially small. For the weighted Taylor remainder the mark is constant on the subsets of each row, so it introduces precisely the same factor.

The unmarked scalar integral is at least c/d , by near-zero positivity of U_d and b_0 . The error in (2) is $o(1/d)$ uniformly, because $\delta \leq M^{-1/3}$ and $d \leq \kappa_0 M$. For the marked scalar integral, use $b_1(z) \leq Cze^{-\theta z}$ and

$$\int_0^\infty ye^{-y}T_d(y)^2 dy = \frac{d+1}{2d}$$

to get $\beta_1(U_d) \leq C/d^2$. When $d \leq \sqrt{M}/\delta$, each term in (3) is $O(1/d^2)$; in particular $\delta \log d = o(1)$ and $d \leq \kappa_0 M$. This proves (4). $\square$

Lemma 3 (Relative-balance improvement). *Hypothesis (1) implies*

$$\max_i u_i \leq \frac{C}{m} \left(s + m^{-1} + m^{-a-1/2} \right).$$

Proof. We first obtain an interval with controlled density ratio. There are fixed $0 < a_0 < b_0$ and a sufficiently small $\kappa > 0$ such that

$$\mu_0(a_0/d < x < b_0/d) \geq c/d, \quad \mu_1(x \leq b_0/d) \leq C/d^2 \quad (3)$$

uniformly for

$$M^{1/4} \leq d \leq D := \min(\kappa M, \sqrt{M}/\delta).$$

To prove the first bound, start with $\mu_0(U_d) \geq c/d$. For $x \leq a_0/d$, the ordinary Gaussian Laguerre bound gives $U_d(x) \leq C$, so this contribution is at most $C(a_0/d + 1/M) \leq C(a_0 + \kappa)/d$ by Lemma 1. For $x \geq b_0/d$, the all-orders estimate at order zero gives $U_d(x) \leq Ce^{\theta x}/(d^2 x^2)$. Since $d\mu_0 \leq Ce^{-x}d\nu$, its contribution is at most

$$\frac{C}{d^2} \int_{b_0/d}^\infty e^{-cx} x^{-2} d\nu \leq C \left(\frac{1}{b_0 d} + \frac{1}{Mb_0^2} \right).$$

Choose b_0 large, then a_0 and κ small. A fixed positive fraction of $\mu_0(U_d)$ remains on the indicated interval, where $U_d \leq C$. This proves its probability lower bound.

For the marked bound use U_e with $e = \lfloor d/(16b_0) \rfloor$. Its near-zero positivity covers $[0, b_0/d]$. For sufficiently large M , $e \geq M^{1/5}$, and the marked comparison applies with $e \asymp d$. Its expectation is $O(1/d^2)$, by (4), proving the other assertion in (3).

At any row with $x \leq a_0/(2d)$, all rows in the interval in (3) lie strictly before that row. Monotonicity and the density relation give

$$(\ell + 2)u_i(q) \frac{c}{d} \leq \mu_1(a_0/d < x < b_0/d) \leq C/d^2, \quad Mu_i(q) \leq C/d.$$

For $x \leq a_0/(4M^{1/4})$ choose an integer d comparable to $\min(D, a_0/(2x))$, with the second argument infinite at zero. It follows that

$$Mu_i \leq C(x + D^{-1}) \leq C(x + M^{-1} + \delta/\sqrt{M}).$$

For larger x , the previous bound (2) suffices, since $\delta \leq M^{-1/3} = o(M^{-1/4}) \leq Cx$. Finally $x \leq s$ and $M \asymp m$. The deleted index was arbitrary, so the asserted improvement holds for every column. $\square$

Proof of Theorem 1. The reviewed relative-balance theorem supplies (1) with $a = 1/3$. Lemma 3 improves its offset to $m^{-5/6}$. A second application improves it to m^{-1} , proving the theorem. Only two applications with fixed constants are involved. $\square$